Curb Management System Technical Report Deliverable 1

1. System Vision document

1. Problem

In busy streets in downtown and Heliopolis, delivery vans, ride hailing cars, buses and cars all share the same parking slots which are limited, and the government has no data on how curb space is used.

2. Vision

The solution for all the companies who suffer from overcrowding and limited parking space is a smart curb management platform that turns each curb into a digital, online bookable space which is time based and has enforced rules using ANPR.

3. Stakeholders

Traffic Planners (design zones and rules), the Head of Traffic Planning (approves rule changes), delivery companies and drivers (book slots), ride-hailing and navigation apps (consume live data), Traffic Police (need legally valid fines), shop owners (customer stops).

4. Key Capabilities

- Zones and rules:
    a. planners draw zones on a map, set rule schedules by day and hour
    b. rules are copyable between streets,
    c. Rule changes need the department head's approval and are published 7 days ahead, except in emergencies.
- Booking:
    a. driver can book 10, 20 or 30 minutes slots in under 30 seconds from a phone
    b. A booking is accepted only if the vehicle fits; otherwise, another zone or time is recommended.
    c. Companies can book recurring slots, capped at 40% of a zone in peak hours.

- Enforcement:

    a. ANPR cameras can detect plates in every zone. Vehicles with no booking or 5+ minutes past will get fined.
    b. A warning first for registered drivers, then fines to the company.

c. Private cars are fined through the traffic violations system. Drivers can dispute with photo evidence.

- Data and services: a live feed for navigation and ride hailing apps. Ride hailing is charged per stop. Fines get sent in batches to a government API every hour.

2. Stakeholder Identification

	Operational Stakeholder	Executive Stakeholder
Internal Stakeholder	**Eng. Rami Fekry**	**Eng. Walid Hussein**
External Stakeholder	**Hassan Abdelfattah**	**Magdy Ghali**

Eng. Rami Fekry is part of the IT department, and he cares about the system's cameras and data platform. That makes him an **internal operational stakeholder**.

Eng. Walid Hussein is the head of traffic planning, and he cares about the traffic flow and clear rules. He won't be interacting with the system directly so he's an **internal executive stakeholder.**

Hassan Abdelfattah works with the delivery companies, and he has interest in the system's slot access and booking. He's an **external operational stakeholder**.

Magdy Ghali is part of the shop owners association. Magdy and his association care about ensuring that their shops remain accessible by customers. They aren't focused on using the system themselves. That makes Magdy an **external executive stakeholder.**

3. Functional Requirments

Cameras and data:

- ANPR cameras should be installed on every curb space.

- The system detects when a vehicle arrives and leaves, and record the exact time of each

- The system sends ANPR readings to the platform in real time, so zone availability is always up to date.

- The system shall receive detection events (plate, time, zone) from ANPR cameras within 2 seconds of a vehicle arriving or departing.

- If a plate read has low confidence, the system sends it to a human operator for manual review before issuing any notices.

- Plate images of vehicles that committed violations are kept for a year, while plate images of vehicles that haven’t committed violations are kept for only 48 hours.

- Data should include a clear photo of the plate, the time, the zone sign visible, and the camera's calibration certificate.

Zones and Rules:

- The system should allow Traffic Planners to divide streets into zones, defining each zone's length in meters, type, and location
- The system should allow Planners to set time-based rules for zones, including allowed use, days, start time, end time, and price
- Disabled parking spaces may not be converted into delivery zones.

- Department Head's approval is required for rule changes, and they should be published 7 days after approval, except in emergencies
- Simulation Tool: The system shall include a simulation environment that allows Traffic Planners to test new proposed rules against past traffic data before applying them to real life streets.

Booking and Registration:

- The system allows delivery companies to register their organization, drivers (with license details), and vehicles (including license plate and vehicle length).
- shop owners can sponsor free 10-minute stops for their customers.
- Drivers can book slots of 10, 20, or 30 minutes in specific zones.
- The system must verify that there is enough free physical length in the zone before accepting a booking.
- If the zone is full, the system will suggest the nearest free zone or time.
- No single company is allowed to hold more than 40% of a zone during peak hours.
- A vehicle cannot book the same zone for more than 60 mins per day.
- Server time is used by the system to validate booking start times and allow a grace period.
- The system should free a booked slot for others if the vehicle never arrives within a set grace period based on ANPR data.
- Electric Cargo Bikes: The system provides smaller and cheaper booking slot options specifically for electric cargo bikes

Violations and Dispute Management:

- The system detects when a space is occupied with no booking and reports it as a possible violation.
- The system detects when a vehicle stays after its booking ends and starts the 5-minute violation timer from the departure data.
- The system will issue a violation if a vehicle has no booking or stays 5 minutes past its booking end time.
- The system issues a warning for a registered company driver’s first violation and then sends fines directly to the company for future violations.
- Companies and drivers are provided with a 7 day window to dispute a fine by uploading photo evidence
- The system allows emergency services to instantly clear a zone which automatically cancels existing bookings and sends notifications to affected drivers.

External API Integrations:

- The system sends private car violations to the government traffic violations API in batches every hour.
- The system provides a live data feed of allowed pickup points to ride-hailing apps, charging them per stop and updating zone availability every minute.

4. Requirements Elicitation Techniques

1. **Interviews:** we have to execute interviews with the technical stakeholders like traffic planning head and smart city IT and the key partners because it's necessary to understand the strategic goals and the technical constraints that we might encounter in the way

2. **Documents Analysis:** we'll gather information and review existing policy drafts, legal emails etc...

   This is important to extract strict unchangeable rules that the system must enforce like for example the 7-day dispute window and the maximum booking percentage.

3. **Questionnaires:** we will distribute digital surveys especially to delivery drivers and car owners in the area of operation Because questionnaires are the most efficient way to gather quantitative feedback and usability requirements from a large group of People who are actively on the road and will use our system.

4. **Stakeholder Workshops:** we will host collaborative sessions combining stakeholders with different priorities, such as traffic planners, shop owners, and delivery fleet managers

Workshops are critical for conflict resolution such as balancing the shop owners' need for customer parking with the delivery companies' need for loading zones.

5. Interview Agendas

Interview Agenda: Internal Executive Stakeholder.

Eng. Walid Hussein

- What metrics/statistics do you need the system to measure or keep track of?
- Do private cars also book zones? If so, is it a good idea to prevent a car from booking a zone for more than an hour per day? Many people such as shop employees certainly will require more than an hour of parking.
- What measures of success will we use to determine whether we should expand to 200 streets or not?
- If you aren't available during emergencies, how should rule changes be handled?

Interview Agenda: Internal Operational Stakeholder.

Eng. Ramy Fekry

- How can we guarantee that the system will be able to withstand the peak 2000 bookings per one hour?
- After we scale from 5 to 200 streets, how can we ensure the system will still withstand peak booking hours? Also, how can we ensure that the system has enough storage to keep violation images for a year?
- At exactly what percent should the threshold be for the system to determine if a plate read is "low confidence" and should be reviewed by a human?

6. Questionnaire

Part A: Drivers

1. What type of driver are you?
2. How long do you usually spend looking for a place to stop in downtown or Heliopolis?
3. Would you use a phone app to book a curb slot before you arrive ?
4. Which slot length fits you best ?
5. Do you agree with a warning first, then a fine if you stay 5+ minutes after your slot ?
6. How do you want to receive reminders ?

Part B: Delivery companies

1. Would you like to same time every day ?
2. Is limiting recurring bookings to 40% of a zone in peak hours acceptable ?
3. If your driver gets a fine after the warning, are you ok with it going to the company ?

Part C: Ride hailing and navigation apps:

1. Would a live feed of free curb zones and rules help your drivers ?
2. Do you agree with paying a fee per stop in a curb zone ?

Part D: Traffic panners and police:

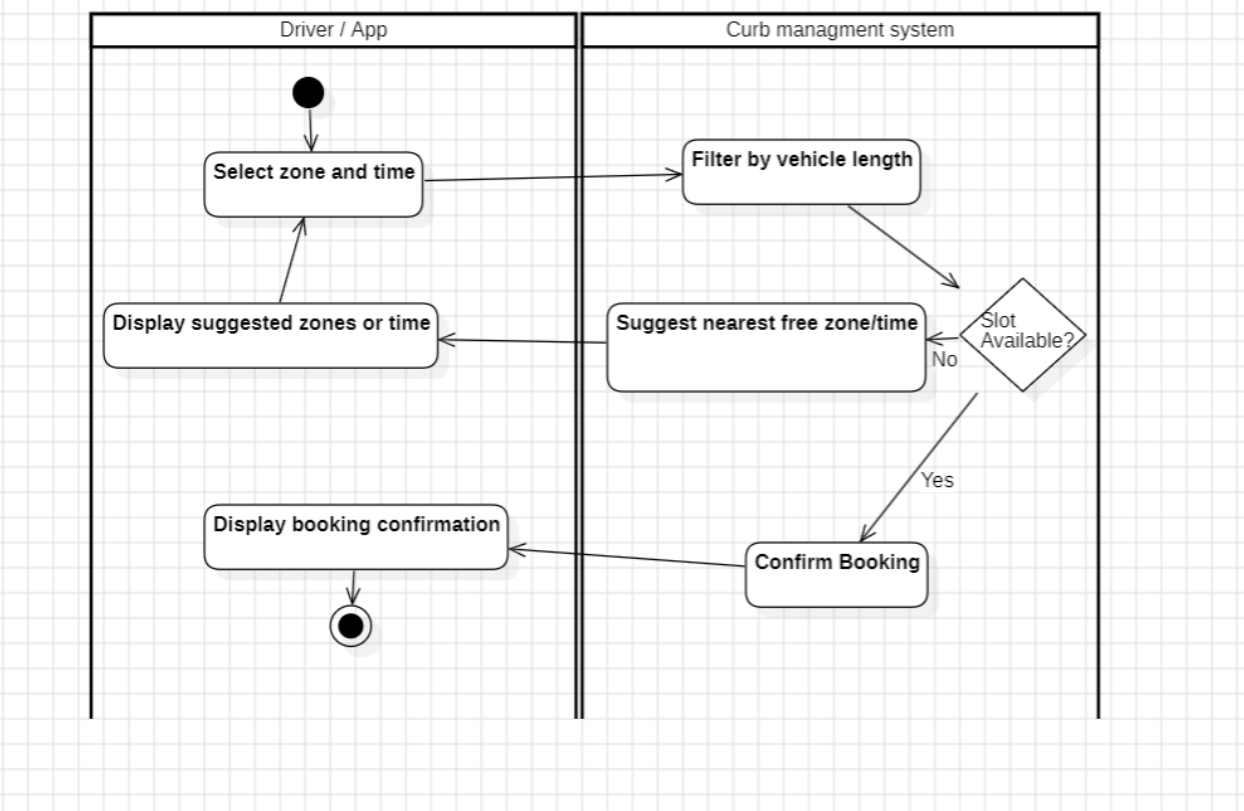
1. Do you currently have data on how curb space is used ?
2. What evidence is needed so a fine is legally valid ?
3. Is publishing rule changes 7 days ahead enough ?

Part E: Shop owners:

1. Do customers have trouble stopping near your shop ?
2. Would short customer-stop slots in front of your shop help you ?

7. Workflow Documentation

Activity diagram A:



Activity Diagram B:

